

SAJAL AGARWAL

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PROFESSIONAL SUMMARY

Results-driven Senior Data Scientist and Generative AI Engineer with 15+ years of experience in machine learning, statistical modeling, advanced analytics, and cloud data engineering. Proven track record deploying production ML models for anomaly detection, fraud detection, and cost optimization. Deep expertise in Python, PostgreSQL, Microsoft Azure (ADF, Databricks, Azure OpenAI), and GenAI frameworks including LangChain, FAISS, RAG pipelines, and GPT-4. Experienced in designing scalable NLP/GenAI solutions and translating complex model outputs into strategic business decisions for C-suite stakeholders. Actively maintaining a public AI/ML project portfolio on GitHub.

TECHNICAL SKILLS

Area	Platforms / Tools
Reporting & BI Tools	Power BI, Tableau, Grafana, Qlik Sense, Advanced Excel, SSRS, MS Access, Cognos TM1
Cloud Platform	Microsoft Azure (Azure Data Factory, Databricks, Azure OpenAI, Azure Functions)
Databases	PostgreSQL, MS SQL Server, IBM Cognos TM1
Programming Tools	Python, FastAPI, Streamlit, Flask, Visual Studio, Azure Data Studio, Databricks
Languages	Python, R, SQL
ML / AI Frameworks	Scikit-learn, XGBoost, Random Forest, Prophet, LangChain, FAISS, RAG Pipelines, GPT-4, HuggingFace
Domains	BFSI, Insurance, Healthcare, Retail, CPG, FMCG, Aviation
Methodology / Tools	Agile, Jira

PROFESSIONAL EXPERIENCE

Data Scientist | Client: Manulife

Tata Consultancy Services — Associate Consultant, Analytics & Insights Jul 2020 – Present | Toronto, ON

- **Developed and deployed an Outlier Detection Model** to trigger alerts for excess Azure charges, generating \$12.8 Million CAD in savings by identifying underutilized cloud resources.
- **Designed and deployed end-to-end ML pipelines** for anomaly detection, fraud detection, and cloud cost optimization, providing technical leadership across the full model lifecycle (development, validation, deployment, monitoring).
- **Built BI dashboards** in Grafana and Power BI using complex PostgreSQL queries to deliver reporting and analytics on Microsoft Azure daily consumption data.
- **Automated Azure billing reporting** via Python scripts and Azure Functions; implemented ETL pipelines in Azure Data Factory for daily data refreshes and database updates.
- **Implemented ensemble ML models** (Random Forest, Gradient Boosting) to enhance fraud detection precision and recall.
- **Designed scalable, maintainable solutions** using Python, FastAPI, Streamlit, and Flask on Databricks for data processing and analytics applications.
- **Applied feature engineering and data transformation** using Pandas, NumPy, and Seaborn within Databricks notebooks for complex analytical workflows.
- **Monitored model performance** through KPI dashboards, presenting actionable insights and recommendations to senior leadership to inform strategic decisions.

Machine Learning Projects:

1. Time-Series Forecasting Model (Facebook Prophet)

- Built a daily time-series forecasting model using Facebook Prophet to predict Azure cloud charges at cost-center level and detect abnormal spending patterns.
- Implemented automated anomaly detection with email alerts to cost-center owners when actual charges exceeded forecasted thresholds.
- Enhanced cloud cost governance by providing resource-level cost breakdowns in alerts, enabling proactive remediation and reducing overall cloud spend.

2. Regression-Based Cloud Cost Forecasting

- Designed a multiple linear regression model to predict monthly Azure cloud charges at cost-center level using resource usage, service type, and historical billing features.
- Integrated forecast outputs into the Azure chargeback and financial reporting process, improving cost visibility and proactive budget management.

3. Fraudulent Loan Application Detection

- Developed an ML-based fraud detection system using borrower credit, income, and loan features; handled class imbalance with SMOTE.
- Achieved F1-score above 0.75 with a Random Forest classifier after hyperparameter tuning via GridSearchCV; identified FICO score, debt-to-income ratio, and loan purpose as top predictors.
- Proposed ensemble model enhancements and a real-time inference architecture for production deployment.

Generative AI Projects:

1. AI Travel Assistant (LangChain + Azure OpenAI + Streamlit)

- Designed a conversational travel assistant providing itinerary planning and flight/hotel suggestions based on natural language prompts.
- Implemented LangChain for LLM orchestration with Azure OpenAI GPT models; integrated Google Places API for real-time data and PostgreSQL for user session history.

2. Natural Language to SQL Query Engine (LangChain + PostgreSQL)

- Built a natural language interface over PostgreSQL databases using LangChain's SQL agent, enabling business users to query structured data without SQL expertise.
- Implemented LangChain create_sql_agent with OpenAI GPT models, SQLAlchemy connections, and conversational memory for multi-turn query sessions.
- Deployed a Streamlit front-end on Databricks for scalable NL-to-SQL inference with dynamic query generation and result visualization.
- GitHub: github.com/Sajalagarwal-ca/langchain_query_NL_PostgreSQL_DB

3. Personal RAG Document Assistant (Streamlit + OpenAI + FAISS)

- Built a local RAG pipeline ingesting multi-format documents (PDF, Word, Excel, CSV, TXT), chunking and embedding content via HuggingFace all-MiniLM-L6-v2, with FAISS vector store for fast semantic retrieval.
- Integrated OpenAI gpt-4o-mini for grounded answer generation; every response includes source file, page, chunk ID, and snippet references.
- Designed a Streamlit UI on LangChain (langchain-classic, langchain-community, langchain-openai) for interactive document Q&A.
- GitHub: github.com/Sajalagarwal-ca/RAG-Document-reader

4. Insurance Fraud Investigation Assistant (RAG-Based Claims Chatbot)

- Developed a GenAI solution using Azure OpenAI GPT-4 to summarize long insurance claim documents and surface anomalies based on historical fraud patterns.
- Combined structured (claims, transaction logs) and unstructured (text, emails, scanned PDFs) data using LangChain and FAISS vector embeddings.
- Built a Power BI front-end dashboard to visualize detected anomalies and generate audit-ready fraud investigation summaries.

Data Scientist | Client: Tata Steel

Tata Consultancy Services — Assistant Consultant, Analytics & Insights Jun 2019 – Jul 2020 | Kolkata, India

- **Identified patterns and applied statistical models** to optimize operational efficiency of the mild steel door production process, improving production yield outcomes.
- **Developed supervised and unsupervised ML models** (regression, clustering, PCA) to analyze and diagnose production process deviations.
- **Designed forecasting and classification models** using multivariate regression, logistic regression, decision trees, and time-series analysis.

- **Experimented with model pruning and quantization techniques** to optimize real-time predictive maintenance models for edge deployment.
- **Developed and maintained KPI dashboards** for cross-functional teams, ensuring data analysis processes aligned with industry standards.

Manager — Analytics

GMR — Delhi International Airport Ltd.

Jul 2018 – Jun 2019 | New Delhi, India

- **Consolidated Management Reports** for all airport terminals for Non-Aero Commercial reporting to CCO/CEO on Qlik Sense and Power BI.
- **Designed data-driven campaigns and promotions** generating INR 2 Million in incremental revenue by increasing sales and transactions across retail and F&B segments.

Deputy Manager — Analytics

WNS Global Services

Dec 2017 – Jul 2018 | Gurugram, India

- **Collaborated with clients and stakeholders** across UK and USA aligned to Healthcare and FMCG analytics verticals.
- **Led a team of 5 analysts** for financial and commercial reporting based on IMS/Nielsen/IRI published data on Power BI.
- **Conducted sales and competitive landscape analysis** to identify high-growth consumer healthcare product segments across SE Asia.
- **Tracked and reported KPIs** (Gross Sales, COGS, Net Sales, Operating Profit, Net Profit) from public financial data for FMCG companies.
- **Generated market insights** through secondary research covering consumer behavior, pricing on digital platforms, and market growth for the Indian FMCG industry.

EDUCATION

Qualification	Institution	Year	Grade
Bachelor of Engineering	Jaypee Institute of Information Technology, Noida	2009	6.8 / 10
MBA	Jaypee Business School	2010	7.3 / 10
Certificate – R Programming	Indian Institute of Technology, Chennai	2017	75%
Diploma in Statistics	Indira Gandhi National Open University	2018	73%

CERTIFICATIONS

- Microsoft Certified: Power BI Data Analyst Associate — Microsoft (2023)
- Data Science Using Python — Udemy (2021)
- Python for Data Science — Udemy (2020)